Table 3. Nicaragua: Public Sector Debt Sustainability Framework—Baseline Scenario 2017–37

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2014	2015	2016	Average [5/]	Standard Deviation [5/]	2017	2018	2019	2020	2021	2022	2017-22 Average	2027	2037	2023-37 Average
Public sector debt 1/	40.2	40.7	41.9			41.5	40.0	39.1	38.2	38.5	38.6		41.0	54.7	
of which: foreign-currency denominated	38.9	39.6	40.8			40.5	39.2	38.4	37.6	38.0	38.2		41.0	54.7	
Change in public sector debt	-2.1	0.5	1.1			-0.4	-1.5	-0.9	-0.9	0.3	0.1		0.6	2.0	
Identified debt-creating flows	-1.3	-0.6	0.9			0.1	-0.5	-0.7	-0.8	0.0	0.0		0.9	2.2	
Primary deficit	1.1	1.1	1.2	-0.1	1.2	1.1	1.1	0.9	0.9	1.6	1.5	1.2	2.1	2.9	2.3
Revenue and grants	26.6	27.4	28.8			28.8	28.8	28.7	28.5	28.3	28.2	28.6	28.2	28.1	28.2
of which: grants	1.1	1.1	0.9			0.9	0.9	0.8	0.8	0.6	0.6	0.8	0.6	0.6	0.6
Primary (noninterest) expenditure	27.7	28.6	30.1			29.9	30.0	29.7	29.4	29.8	29.8	29.8	30.3	31.0	30.5
Automatic debt dynamics	-2.3	-1.7	-0.4			-1.0	-1.6	-1.6	-1.7	-1.6	-1.5		-1.2	-0.6	
Contribution from interest rate/growth differential	-1.8	-1.2	-1.2			-1.5	-1.6	-1.6	-1.6	-1.6	-1.4		-1.2	-0.6	
of which: contribution from average real interest rate	0.2	0.6	0.6			0.4	0.1	0.1	0.1	0.1	0.2		0.6	1.6	
of which: contribution from real GDP growth	-1.9	-1.9	-1.8			-1.8	-1.7	-1.7	-1.7	-1.6	-1.7		-1.7	-2.3	
Contribution from real exchange rate depreciation	-0.5	-0.5	0.8			0.5	-0.1	0.0	-0.1	0.0	0.0		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-0.8	1.1	0.3			-0.5	-1.0	-0.2	-0.1	0.3	0.0		-0.3	-0.2	
Other Sustainability Indicators															
PV of public sector debt	...	...	31.6			30.9	29.3	28.4	27.9	28.6	28.7		31.1	46.8	
of which: foreign-currency denominated	...	...	30.5			30.0	28.5	27.7	27.4	28.1	28.3		31.1	46.8	
of which: external	...	...	22.1			22.9	23.5	24.2	25.1	26.1	26.7		28.6	29.9	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	6.0	5.8	5.1			5.8	4.7	3.8	3.4	3.8	3.8		5.0	9.7	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	109.6			107.3	101.8	99.0	97.9	101.0	101.5		110.4	166.1	
PV of public sector debt-to-revenue ratio (in percent)	...	...	113.3			110.8	105.1	101.9	100.6	103.4	103.9		113.0	170.0	
of which: external 3/	...	...	79.3			82.1	84.4	86.8	90.4	94.5	96.6		103.7	108.9	
Debt service-to-revenue and grants ratio (in percent) 4/	9.2	9.6	6.8			10.9	10.0	9.3	8.2	7.8	8.0		10.5	24.1	
Debt service-to-revenue ratio (in percent) 4/	9.6	10.0	7.0			11.2	10.3	9.5	8.4	7.9	8.2		10.8	24.7	
Primary deficit that stabilizes the debt-to-GDP ratio	3.2	0.6	0.1			1.5	2.7	1.8	1.8	1.3	1.5		1.5	0.9	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	4.8	4.9	4.7	4.2	2.8	4.5	4.3	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5
Average nominal interest rate on forex debt (in percent)	2.3	2.9	2.9	3.0	0.6	3.0	2.6	2.4	2.2	2.1	2.7	2.5	3.5	5.3	4.0
Average real interest rate on domestic debt (in percent)	-2.8	-2.0	1.3	-2.3	3.5	-0.4	-1.9	-1.5	-1.7	-1.6	-0.8	-1.3	-1.6	...	-1.6
Real exchange rate depreciation (in percent, + indicates depreciation	-1.4	-1.2	2.2	-1.0	2.9	1.2	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	8.4	7.5	4.1	7.9	3.6	5.9	7.4	7.1	7.3	7.2	7.2	7.0	7.1	7.1	7.1
Growth of real primary spending (deflated by GDP deflator, in percer	5.6	8.3	10.2	2.4	4.0	3.9	4.5	3.5	3.7	5.9	4.2	4.3	4.7	4.1	4.8
Grant element of new external borrowing (in percent)	...	...	...	...	...	34.2	33.6	33.0	25.3	24.6	26.9	29.6	19.0	14.2	...

Sources: Country authorities; and staff estimates and projections.

1/ Consolidated public sector

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Nicaragua: Sensitivity Analysis for Key Indicators of Public Debt 2017–37

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of Debt-to-GDP Ratio								
Baseline	31	29	28	28	29	29	31	47
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	31	28	26	25	24	23	17	11
A2. Primary balance is unchanged from 2017	31	29	29	28	28	28	27	30
A3. Permanently lower GDP growth 1/	31	30	29	29	31	32	41	84
A4. Alternative Scenario : ALBA debt contingent liability	31	46	45	44	45	45	47	61
A5. Adverse Scenario	31	31	32	33	36	38	49	80
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	31	31	33	34	36	37	47	76
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	31	29	29	28	29	29	31	47
B3. Combination of B1-B2 using one half standard deviation shocks	31	29	28	29	30	31	38	61
B4. One-time 30 percent real depreciation in 2018	31	43	41	40	40	40	43	69
B5. 10 percent of GDP increase in other debt-creating flows in 2018	31	39	38	37	38	38	39	54
PV of Debt-to-Revenue Ratio 2/								
Baseline	107	102	99	98	101	101	110	166
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	107	98	92	88	86	81	60	40
A2. Primary balance is unchanged from 2017	107	102	99	99	100	99	95	106
A3. Permanently lower GDP growth 1/	107	103	102	103	109	113	147	298
A4. Alternative Scenario : ALBA debt contingent liability	107	160	156	154	158	159	167	217
A5. Adverse Scenario	107	107	111	116	126	134	173	285
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	107	107	113	118	127	132	167	271
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	107	102	99	98	101	102	111	166
B3. Combination of B1-B2 using one half standard deviation shocks	107	101	98	100	106	110	133	215
B4. One-time 30 percent real depreciation in 2018	107	148	143	141	143	142	153	244
B5. 10 percent of GDP increase in other debt-creating flows in 2018	107	135	133	131	134	134	140	191
Debt Service-to-Revenue Ratio 2/								
Baseline	11	10	9	8	8	8	10	24
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	11	10	9	5	4	2	-6	-14
A2. Primary balance is unchanged from 2017	11	10	9	8	8	8	6	6
A3. Permanently lower GDP growth 1/	11	10	9	9	9	10	20	60
A4. Alternative Scenario : ALBA debt contingent liability	11	10	11	9	11	12	14	28
A5. Adverse Scenario	11	10	10	9	9	10	16	38
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	11	10	10	11	13	16	27	53
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	11	10	9	8	8	8	11	24
B3. Combination of B1-B2 using one half standard deviation shocks	11	10	9	7	7	9	17	38
B4. One-time 30 percent real depreciation in 2018	11	12	12	13	13	15	24	53
B5. 10 percent of GDP increase in other debt-creating flows in 2018	11	10	13	34	13	27	17	31

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

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